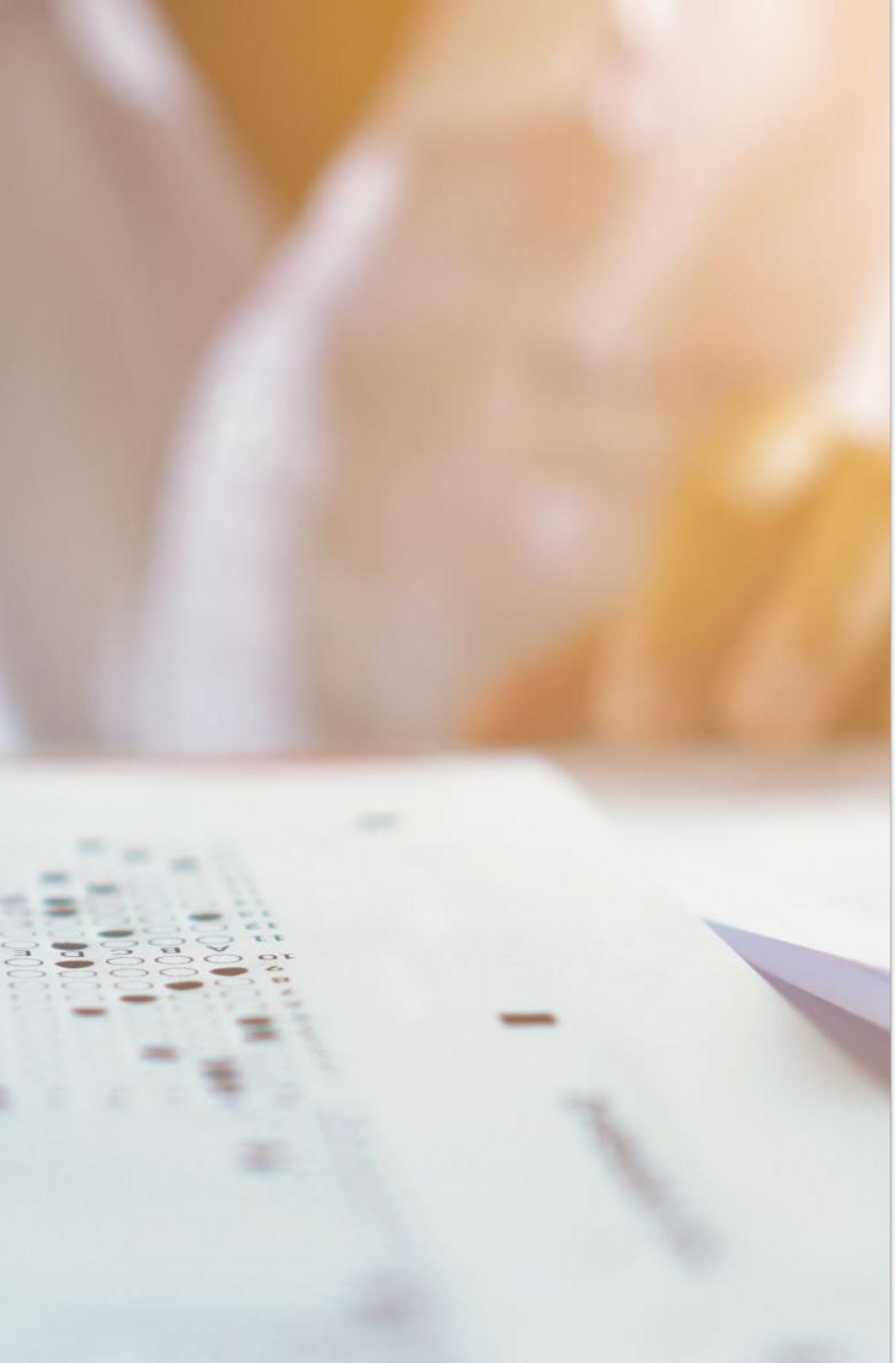

GROUP – A

DIGITAL SIGNING REQUIREMENT IN MSIX

- Purpose:
- Authenticity (trusted source)
- Integrity (no tampering)
- Mandatory for Windows 10/11 installations, including sideloading policies.
- Certificate types:
- Self-signed: Testing only
- CA-signed: Production
- Tip: Check certificate validity to prevent rejection.

DIGITAL SIGNING REQUIREMENTS

- A code-signing certificate (one of the following):
  - Self-Signed Certificate → for testing/sideloading only.
  - Enterprise CA Certificate → for internal corporate deployments (via Intune, SCCM).
  - Public CA Certificate → required if distributing via Microsoft Store or for external customers.
- Supported signature algorithm → SHA256.
- Timestamping strongly recommended so signature remains valid even after certificate expiry.

DIGITAL SIGNING IMPLEMENTATION STEPS

1. Obtain CA-signed certificate

2. Package application into MSIX

3. Sign using PowerShell or Sign Tool

4. Verify signature with Get-Appx PackageManifest

Best Practices:

- Sign after packaging

- Monitor certificate expiry

- Test in pilot environment.

TROUBLESHOOTING MSIX PACKAGES

- Common Errors:

- Signature mismatch or invalid certificate

- Missing dependencies

- Package fails to install

- Tools & Logs:

- PowerShell Verbose Mode

- Event Viewer: AppXDeployment-Server logs

- ProcMon Trace file & registry access

- Tips:

- Ensure Windows is updated

- Validate dependencies before packaging.

TROUBLESHOOTING TOOLS

Event Viewer – Check MSIX installation, deployment, and runtime logs.

PowerShell Cmdlets – Commands to test, repair, and debug MSIX apps.

MSIX Packaging Tool – Used for creating, editing, and validating MSIX packages.

Process Monitor (ProcMon) – Tracks file system, registry, and process activity.

Dependency Walker – Identifies missing DLLs or dependencies.

App Certification Kit – Validates MSIX packages for compatibility and Store submission.

PREPARING THE MANAGEMENT INFRASTRUCTURE (SCCM)

- Install and Configure SCCM
- Database Setup
- Networking & Boundaries
- Site System Roles
- Client Deployment
- Security & Permissions
- Monitoring & Reporting

SCCM ROLES AND CONSOLE OVERVIEW

- Primary Roles:
- - Site Server
- -Site database
- - Management Point
- - Distribution Point
- SCCM Console:
- - Deploy applications
- - Manage clients
- - Monitor inventory & compliance

PLANNING & BEST PRACTICES

- Plan device count and deployment
- Test in pilot collections
- Optimize bandwidth
- Keep SCCM & SQL updated
- Document configurations & boundaries

SCCM CLIENT OVERVIEW

INSTALLED ON ENDPOINTS

CAPABILITIES:

- HARDWARE & SOFTWARE INVENTORY

- REMOTE TROUBLESHOOTING

- COMPLIANCE REPORTING

- SOFTWARE DEPLOYMENT

DEPLOYMENT METHODS: CLIENT PUSH, GPO, MANUAL, SCRIPTED

INSTALLING & CONFIGURING SCCM CLIENT

- Steps:
- 1. Deploy client via SCCM console
- 2. Configure inventory, updates, power management
- 3. Verify installation via Control Panel
- Logs: ccmsetup.log, ClientIDManagerStartup.log
- Tip: Keep clients updated to match site version.

CLIENT MANAGEMENT TASKS

- Trigger hardware/software inventory cycles
- Enforce compliance policies
- Remote actions: Remote Control, PowerShell scripts, Restart
- Monitor client health: SCCM Console → Monitoring → Client Status
- Tip: Automate inventory cycles for regular reporting.

INVENTORY MANAGEMENT FOR PCS & APPLICATIONS

HARDWARE INVENTORY: CPU, RAM, DISK, NETWORK ADAPTERS

SOFTWARE INVENTORY: INSTALLED APPS, UPDATES, VERSIONS

BENEFITS: LICENSE COMPLIANCE, ASSET MANAGEMENT, DEPLOYMENT PLANNING

EXAMPLE: IDENTIFY OUTDATED SOFTWARE BEFORE UPDATES.